

1. What is the output of the program below? (10%)

```
int main() {
    bool k = 7;
    for (int i = 8; i <= 12; i++) {
        if (i % 2 == 0) {
            continue;
        }
        for (int j = 3; j <= 9; j++) {
            if (j > 6) {
                break;
            }
            cout << (double)j/i << ", " << (double)(i/j) << endl;
        }
    }
    cout << (int)k << endl;
}
```

2. Write a simple C++ program to demonstrate how "**short circuit evaluation**" works. You should also explain your program. (10%)
3. Explain what is **call stack**. In addition, please describe the advantage and disadvantage of **inline function**. (10%)
4. Write a simple C++ program to demonstrate "**function overloading**". You should also explain your program. (10%)
5. Write a simple C++ program to demonstrate "**function template**". You should also explain your program. (10%)
6. What is the output of the following program? (5%) What problem does the following program have? You should explain the problem in detail. (5%)

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     int p = 8;
5     int * q = new int;
6     int * r = new int;
7     *q = p;
8     *r = 6;
9     r = q;
10    *r += *q;
11    cout << *q << ", " << *r << endl;
12}
```

7. Explain what the difference between the **smart pointer** and the **raw pointer** is. (10%)
8. What is the output of the program below? (10%)

```
int * f(int c[3]) {
    c[2] = 6;
    return c;
}

int main() {
    int a[3] = {5, 2, 3};
    int *b;
    b = f(a);
    b[1] = a[0]+a[2];
    a[0] = *(b+1);
    for (int x: a) {
        cout << x << endl;
    }
}
```

9. Write three programs to demonstrate Call by Value, Call by Address, and Call by Reference in C++. You should explain the programs you write. (10%)

10.(1) Figure out the lines that will cause compilation error. (2) After removing the lines that will cause compilation error, what is the output of this program? (10%)

```
1 #include<iostream>
2 using namespace std;
3 class Midterm {
4 private:
5     int i1;
6     static int si1;
7 public:
8     int i2;
9     Midterm(int v) {
10         i1 = v+1;
11         i2 = v+2;
12         si1 = v+3;
13     }
14     static int si2;
15     static void setVariables(int x) {
16         i1 = x+1;
17         si2 = x+2;
18     }
19 };
20
21 int Midterm::si1 = 6;
22 int Midterm::si2 = 8;
23
24 int main(int argc, const char * argv[]) {
25
26     Midterm midterm1(1);
27     Midterm midterm2(6);
28     midterm1.setVariables(8);
29
30     cout << Midterm::si1 << endl;
31     cout << Midterm::si2 << endl;
32
33     midterm1.si2 = 7;
34
35     cout << midterm1.i1 << endl;
36     cout << midterm2.i1 << endl;
37     cout << midterm1.i2 << endl;
38     cout << midterm2.i2 << endl;
39     cout << midterm1.si1 << endl;
40     cout << midterm2.si1 << endl;
41     cout << midterm1.si2 << endl;
42     cout << midterm2.si2 << endl;
43 }
```